Prac: Properties of common substanes

Name: _____

Aim: To investigate the properties of common ionic, metallic and covalent substances.

Equipment and Materials:
Material samples:
- Aluminium foil
- Calcium carbonate (powder and chips)
- Copper sulfate
- Copper wire
- Magnesium ribbon
- Methylated spirits
- Sodium chloride
- Sugar
- Water

Equipment:
- Power pack
- Lamp
- Wires
- 50mL beaker

Method:
Test each material sample with the following tests and complete the results table.
1. Does it bend?
2. Does it conduct electricity?
3. Does it dissolve in water?
4. If it dissolves, does it conduct electricity?

Results:

Material	Bend?	Conduct?	Dissolve?	Solution conducts?
Aluminium foil				
Calcium carbonate				
Copper sulfate				
Copper wire				
Magnesium ribbon				
Methylated spirits				
Sodium chloride				
Sugar				
Water				

Discussion:

For each material, suggest whether it is ionic, metallic, or covalent. Include a reason.

Material	Bond type	Reason
Aluminium foil		
Calcium carbonate		
Copper sulfate		

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Copper wire		
Magnesium ribbon		
Methylated spirits		
Sodium chloride		
Sugar		
Water		

Conclusion:

Write one sentence that relates to the aim.

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